

Math 103

Class 8

1. Mean, median, mode, variance and standard deviation for grouped data
2. Review

$$\text{Mean} = \bar{x} = \frac{\sum x}{n}$$

Mode = score that occurs most often.

Median = MD = middle number

$$\text{Variance} = S^2 = \frac{\sum (x - \bar{x})^2}{n-1} \quad \text{or} \quad S^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}$$

Round decimal answers to one more place than the data.

The "variance" tells one how "compact" the data is.
It tells how far away from the mean the data is.

Standard Deviation = $\sqrt{s^2} = s$

Empirical Rule: About 2/3 of all the scores will be within one standard deviation of the mean.

Virtually all of the scores are within three standard deviations of the mean.

3. To find the mean for "grouped data"

$$\bar{x} = \frac{\sum f \cdot x_m}{n} \quad x_m = \text{class midpoint}$$

Class Limits	Class Boundaries	f	x_m	$f \cdot x_m$	Cumulative frequency	$\bar{x} = \frac{\sum f \cdot x_m}{n}$
1-3	.5 - 3.5	3	2	6	3	= 172/20 = 8.6
4-6	3.5 - 6.5	5	5	25	8	
7-9	6.5 - 9.5	4	8	32	12	
10-12	9.5 - 12.5	3	11	33	15	mode = 5
13-15	12.5 - 15.5	3	14	42	18	
16-18	15.5 - 18.5	2	17	34	20	
		20		172		

The mode for grouped data is the class midpoint of the class

with the highest frequency.

2.

To find the median for grouped data: $MD = \frac{(n/2) - cf(w)}{f} + L_m$

$$\begin{aligned} n &= 20 && \text{(sum of frequencies)} \\ cf &= 8 && \text{(cum. freq. of class immediately preceding median class)} \\ &&& \text{In this problem, 7-9 is the median class} \\ f &= 4 && \text{(frequency of median class)} \\ w &= 9.5-6.5 && \text{(width of median class)} \\ &= 3 \\ L_m &= 6.5 && \text{(lower boundary of median class)} \\ MD &= \frac{(20/2) - 8(3)}{4} + 6.5 \\ &= 8.0 \end{aligned}$$

Fill in the x_m^2 and the $f \bullet x_m^2$ columns

<u>x_m^2</u>	<u>$f \bullet x_m^2$</u>
4	12
25	125
64	256
121	363
196	588
289	<u>578</u>
	1922

Calculate the variance and standard deviation:

$$\begin{aligned} s^2 &= \frac{\Sigma f \bullet x_m^2 - (\Sigma f \bullet x_m)^2 / n}{n - 1} \\ &= \frac{1922 - (172^2) / 20}{19} \\ &= \frac{1922 - 29584 / 20}{19} \\ &= \frac{1922 - 1479.2}{19} \\ &= \frac{442.8}{19} \\ &= 23.305 \end{aligned}$$

$$\approx 23.3$$

$$\text{Therefore } s = 4.8276$$

$$\approx 4.8$$

same answer with sq. rt of 23.3

3.

6. In a class of 29 students, this distribution of quiz scores was recorded:

<u>Grade</u>	<u>Frequency</u>
0-2	1
3-5	3
6-8	5
9-11	14
12-14	6

Find the mean, median and mode for the grouped data.

Class		Cumulative			
Grade	Boundaries	f	x_m	$f \bullet x_m$	Frequency
0-2	-.5-2.5	1	1	1	1
3-5	2.5-5.5	3	4	12	4
6-8	5.5-8.5	5	7	35	9
9-11	8.5-11.5	14	10	140	23
12-14	11.5-14.5	<u>6</u>	13	<u>78</u>	29
		29		266	

$$\begin{aligned} \bar{x} &= \frac{\sum f \bullet x_m}{n} \\ &= \frac{266}{29} \\ &= 9.1724 \\ &\approx 9.2 \end{aligned}$$

mode = 10

median class = 9-11

$$\begin{aligned} MD &= \frac{(n/2) - cf(w)}{f} + L_m \\ &= \frac{(29/2) - 9(3)}{14} + 8.5 \\ &= \frac{14.5 - 9(3)}{14} + 8.5 \\ &= \frac{5.5(3)}{14} + 8.5 \\ &= 1.17857 + 8.5 \\ &= 1.179 + 8.5 \\ &= 9.679 \\ &\approx 9.7 \end{aligned}$$

Calculation of the variance and standard deviation

$$s^2 = \frac{2708 - (266^2) / 29}{28}$$

$$= \frac{2708 - (70756/29)}{28}$$

$$= \frac{2708 - 2439.862}{28}$$

$$= \frac{268.138}{28}$$

$$= 9.576$$

Therefore the st. dev. is:

$$s = \sqrt{9.576}$$

$$\approx 3.095$$

$$\approx 3.1$$